Practical 1: Generate a First RetailPulse Report

Related topic: Introduction to IBM Cognos Analytics – Reporting

OBJECTIVE

Navigate the Cognos Analytics Reporting environment and generate a running list report against the RetailPulse Sales package.

MATERIALS REQUIRED

IBM Cognos Analytics access, RetailPulse Sales package, and a personal content folder.

STEP-BY-STEP PROCEDURE

1. Open IBM Cognos Analytics and select New > Report.
2. Choose the RetailPulse Sales package as the data source.
3. Choose the List template.
4. From the Source tab, drag Region, Product Category, and Sales Amount onto the canvas.
5. Click Run to preview the report.
6. Save it as “RetailPulse – First Sales Report” in your personal folder.

SAMPLE SOLUTION / EXPECTED OUTPUT

The report should contain three columns: Region, Product Category, and Sales Amount. Example layout:

Region	Product Category	Sales Amount
North	Electronics	(live value)
South	Apparel	(live value)
West	Grocery	(live value)

EXPECTED OUTCOME

The report displays live RetailPulse rows, with correctly labelled columns, and is saved and reopened successfully.

VIVA-VOCE QUESTIONS WITH ANSWERS

Q. What is the main difference between Cognos Reporting and dashboards?
Answer: Reports are structured and formatted for detailed data; dashboards emphasize interactive visual summaries.

Q. Name four tabs/areas available in the Reporting side panel.
Answer: Common areas include Source, Data, Toolbox, and Pages; labels may vary by version.

Q. Which template is suitable for a pivot-style summary?
Answer: Crosstab, because it summarizes measures across row and column categories.

Q. What must be selected before a report can display data?
Answer: A data source or package.

Q. Where can you add or rename a report page?
Answer: In the Pages area/pane.

Record note: The output examples are illustrative layouts. Run the practical with the supplied RetailPulse data to obtain your actual values.

Practical 2: Build a Grouped RetailPulse Sales List Report

Related topic: Create List Reports

OBJECTIVE

Construct a list report, group it by Region and Store Name, format columns, and add a summarizing footer.

MATERIALS REQUIRED

IBM Cognos Analytics access and RetailPulse Sales package; Practical 1 may be reused.

STEP-BY-STEP PROCEDURE

1. Create a new List report using the RetailPulse Sales package.
2. Add Region, Store Name, Product Category, Quantity, and Sales Amount.
3. Group by Region, then Store Name.
4. Format Sales Amount as currency and right-align numeric columns.
5. Add a list footer under Sales Amount and select Total (sum).
6. Add a page header with report title and run date; run and save.

SAMPLE SOLUTION / EXPECTED OUTPUT

Grouped list example:

Region	Store Name	Product Category	Quantity	Sales Amount
North	Store A	Electronics	…	■…
	Store A	Apparel	…	■…
	Region total		Σ	Σ Sales
South	Store B	Grocery	…	■…
Grand total			Σ	Σ Sales

EXPECTED OUTCOME

Rows are grouped by region and store; currency formatting, totals, and page header appear correctly.

VIVA-VOCE QUESTIONS WITH ANSWERS

Q. What happens when a column is grouped?
Answer: Repeated group values are displayed as group headings rather than repeated for every detail row.

Q. Why group Region before Store Name?
Answer: It creates a hierarchy of regions containing their stores.

Q. How do you format Sales Amount as currency?
Answer: Select the column and choose currency formatting in its properties.

Q. Name two aggregate functions besides Total.
Answer: Average and Count; Minimum and Maximum are also common.

Q. How is a page footer different from a list footer?
Answer: A page footer appears at the bottom of each page; a list footer belongs to the list and can show aggregate values.

Record note: The output examples are illustrative layouts. Run the practical with the supplied RetailPulse data to obtain your actual values.

Practical 3: Filter RetailPulse Sales by Region and Date

Related topic: Focus Reports Using Filters

OBJECTIVE

Apply a simple Region filter and an advanced date filter, then convert the Region condition into a prompt.

MATERIALS REQUIRED

IBM Cognos Analytics access and the grouped RetailPulse list report from Practical 2.

STEP-BY-STEP PROCEDURE

1. Open the grouped RetailPulse list report.
2. Open the Filters pane and add Region = “North”.
3. Add an advanced detail filter for Sales Date between the quarter start and quarter end.
4. Combine the Region and date conditions with AND.
5. Run the report and check that only matching rows remain.
6. Convert the Region filter into a prompted filter so another region can be selected when running the report.

SAMPLE SOLUTION / EXPECTED OUTPUT

Filter logic: Region = “North” AND Sales Date is within the chosen quarter. The sample table below shows the expected filtered structure, not actual live values.

Region	Sales Date	Product Category	Sales Amount
North	Within selected quarter	Electronics	(filtered value)
North	Within selected quarter	Apparel	(filtered value)
Only North + date range			

EXPECTED OUTCOME

Only rows matching both conditions remain. Changing the prompt to South should return South-region rows within the selected date range.

VIVA-VOCE QUESTIONS WITH ANSWERS

Q. Detail filter vs summary filter?
Answer: A detail filter filters individual records; a summary filter filters grouped/aggregated results.

Q. When is an advanced detail filter useful?
Answer: When multiple conditions or a more specific expression is required.

Q. Which operator requires both conditions to be true?
Answer: AND.

Q. Why use a prompted filter?
Answer: It lets report viewers choose a value instead of using a fixed hard-coded value.

Q. How can filtering affect performance?
Answer: It may reduce the amount of data processed and displayed, depending on the source and query.

Record note: The output examples are illustrative layouts. Run the practical with the supplied RetailPulse data to obtain your actual values.

Practical 4: Build a RetailPulse Region × Product Crosstab

Related topic: Create Crosstab Reports

OBJECTIVE

Summarize RetailPulse sales across Region and Product Category using Sales Amount and Quantity, including row and column totals.

MATERIALS REQUIRED

IBM Cognos Analytics access and RetailPulse Sales package.

STEP-BY-STEP PROCEDURE

1. Start a new report and choose the Crosstab template.
2. Drag Region into the row edge and Product Category into the column edge.
3. Drag Sales Amount into the crosstab body.
4. Add Quantity as a second measure.
5. Turn on row and column summary totals.
6. Format Sales Amount as currency, run the report, and save it as “RetailPulse Region-Product Crosstab”.

SAMPLE SOLUTION / EXPECTED OUTPUT

Illustrative crosstab structure:

Region	Electronics Sales / Qty	Apparel Sales / Qty	Grocery Sales / Qty	Total
East	Σ Sales / Σ Qty	Σ Sales / Σ Qty	Σ Sales / Σ Qty	Σ
North	Σ Sales / Σ Qty	Σ Sales / Σ Qty	Σ Sales / Σ Qty	Σ
South	Σ Sales / Σ Qty	Σ Sales / Σ Qty	Σ Sales / Σ Qty	Σ
West	Σ Sales / Σ Qty	Σ Sales / Σ Qty	Σ Sales / Σ Qty	Σ
Column total	Σ Sales / Σ Qty	Σ Sales / Σ Qty	Σ Sales / Σ Qty	Grand total

EXPECTED OUTCOME

The crosstab aligns regions with product categories; each cell aggregates Sales Amount and Quantity, and row/column totals reconcile with the grouped list report when run on the same data.

VIVA-VOCE QUESTIONS WITH ANSWERS

Q. How does a crosstab differ from a list report?
Answer: A list displays records in rows; a crosstab summarizes measures at intersections of row and column categories.

Q. Which three drop zones define a basic crosstab?
Answer: Rows, columns, and measures/body.

Q. How do you add a second measure?
Answer: Drag the second measure, such as Quantity, into the crosstab body beside the first measure.

Q. Name two data source types for Cognos reports.
Answer: A package/data module and a supported database or uploaded data source.

Q. What do row and column summaries add?
Answer: Totals for each row/column and grand totals.

Record note: The output examples are illustrative layouts. Run the practical with the supplied RetailPulse data to obtain your actual values.

Practical 5: Build the RetailPulse Sales Prompt Report

Related topic: Design a Cognos Prompt Report for Sales Analysis

OBJECTIVE

Design a prompt page with Region, date-range, and cascading Store prompts and connect each control to a report filter.

MATERIALS REQUIRED

IBM Cognos Analytics access and the Region-Product Crosstab from Practical 4.

STEP-BY-STEP PROCEDURE

1. Open the saved Region-Product Crosstab.
2. Insert a prompt page before the crosstab page.
3. Add a Value prompt for Region and a Date-range prompt.
4. Add a cascading Store prompt that updates based on Region.
5. Link each prompt to its matching report filter.
6. Set default values, add a Finish button, then run and test with more than one region.

SAMPLE SOLUTION / EXPECTED OUTPUT

Expected prompt-page mock-up:

Prompt	Example selection
Region (value prompt)	North ▼
Reporting period (date range)	Start date — End date
Store (cascading prompt)	Stores belonging to North ▼
Action	[Finish]

EXPECTED OUTCOME

Selecting North and a date range should return only matching rows. Changing Region to South should update the Store choices and crosstab results.

VIVA-VOCE QUESTIONS WITH ANSWERS

Q. What problem do prompts solve?
Answer: They let viewers choose report criteria at run time rather than relying on fixed filters.

Q. When is a cascading prompt useful?
Answer: When the second list depends on the first selection, such as stores available within a selected region.

Q. Why set default values?
Answer: To provide sensible initial criteria and make the report easier to run.

Q. Which element applies the selections to data?
Answer: The report filters linked to the prompt controls.

Q. How would you test multiple regions?
Answer: Run with North, then South, and verify the displayed rows and store options change correctly.

Record note: The output examples are illustrative layouts. Run the practical with the supplied RetailPulse data to obtain your actual values.

Practical 6: Build the RetailPulse Customer Insights Dashboard

Related topic: Dashboard for Customer Insights Visualization

OBJECTIVE

Assemble a dashboard with a summary card, new-versus-returning customer trend, loyalty-segment breakdown, and shared Region filter.

MATERIALS REQUIRED

IBM Cognos Analytics access and RetailPulse Customers data module.

STEP-BY-STEP PROCEDURE

1. Create a new Dashboard and connect the RetailPulse Customers data module.
2. Add a summary card for total active customers.
3. Add a line chart tracking new versus returning customers over time.
4. Add a bar chart showing customers by loyalty segment.
5. Add a Region filter widget and test that it cross-filters the other widgets.
6. Arrange the widgets clearly and save the dashboard.

SAMPLE SOLUTION / EXPECTED OUTPUT

Illustrative dashboard layout:

RetailPulse Customer Insights Dashboard	
■■■■■■■■■■■■■■■■■■■■■■■	■■■■■■■■■■■■■■■■■■■

Practical 7: Import and Transform RetailPulse Data in Power BI

Related topic: Power BI Interface, Data Import and Transformation

OBJECTIVE

Import the RetailPulse workbook, explore Power BI Desktop, and clean/shape data using Power Query.

MATERIALS REQUIRED

Power BI Desktop and RetailPulse_Sales.xlsx containing Sales, Product, and Store tables.

STEP-BY-STEP PROCEDURE

1. Open Power BI Desktop and identify Report, Table, and Model views plus the ribbon and panes.
2. Choose Home > Get data > Excel workbook; select RetailPulse_Sales.xlsx.
3. In Navigator, select Sales, Product, and Store, then click Transform Data.
4. Set Sales Date to Date, Quantity to Whole Number, and Sales Amount to Fixed decimal number.
5. Remove blank and duplicate Sales rows, replace null Quantity values with 0, and rename unclear columns.
6. Add conditional column Sale Size: Large when Sales Amount ≥ 1000; otherwise Standard.
7. Review Applied Steps, click Close & Apply, and save as RetailPulse_Analysis.pbix.

SAMPLE SOLUTION / EXPECTED OUTPUT

Applied Steps example and sample cleaned row:

Applied step	What it does
Source	Connects to RetailPulse_Sales.xlsx
Navigation	Selects Sales table
Changed Type	Sets Date, Whole Number, Fixed decimal
Removed Blank Rows	Removes empty rows
Removed Duplicates	Removes duplicate records
Replaced Value	Changes null Quantity to 0
Added Conditional Column	Creates Sale Size
Output example	OrderID | Sales Date | Quantity | Sales Amount | Sale Size
Sample row	ORD1132 | 2024-01-09 | 3 | 2675.33 | Large

EXPECTED OUTCOME

All three tables load without errors; data types are correct; blank/duplicate rows are removed; Sale Size labels each row; changes are recorded in Applied Steps.

VIVA-VOCE QUESTIONS WITH ANSWERS

Q. Import vs DirectQuery?
Answer: Import loads a copy of data into the model; DirectQuery sends queries to the source as visuals are used.

Q. What are Applied Steps?
Answer: A recorded sequence of Power Query transformations that can be reviewed and rerun on refresh.

Q. Which view checks relationships?
Answer: Model view.

Q. Why set data types correctly?
Answer: Correct types support valid calculations, sorting, filtering, and visuals.

Q. Transform Data vs Close & Apply?
Answer: Transform Data opens Power Query; Close & Apply loads the applied transformations into the model.

Record note: The output examples are illustrative layouts. Run the practical with the supplied RetailPulse data to obtain your actual values.

Practical 8: Model, Calculate, and Publish a RetailPulse Sales Report

Related topic: Relationships, Basic DAX, Visualizations and Publishing

OBJECTIVE

Create table relationships, DAX measures and a calculated column, build interactive visuals, and publish the report.

MATERIALS REQUIRED

Power BI Desktop, RetailPulse_Analysis.pbix from Practical 7, and a Power BI service account for publishing.

STEP-BY-STEP PROCEDURE

1. In Model view, create one-to-many relationships Product[ProductID] → Sales[ProductID] and Store[StoreID] → Sales[StoreID].

2. Create measures for Total Sales, Total Quantity, and Average Order Value.

3. Create calculated column Sales per Unit using Sales Amount divided by Quantity.

4. Build a page with a Total Sales card, Sales by Product Category bar chart, Sales by month line chart, and Region slicer.

5. Test cross-filtering and add clear titles and axis labels.

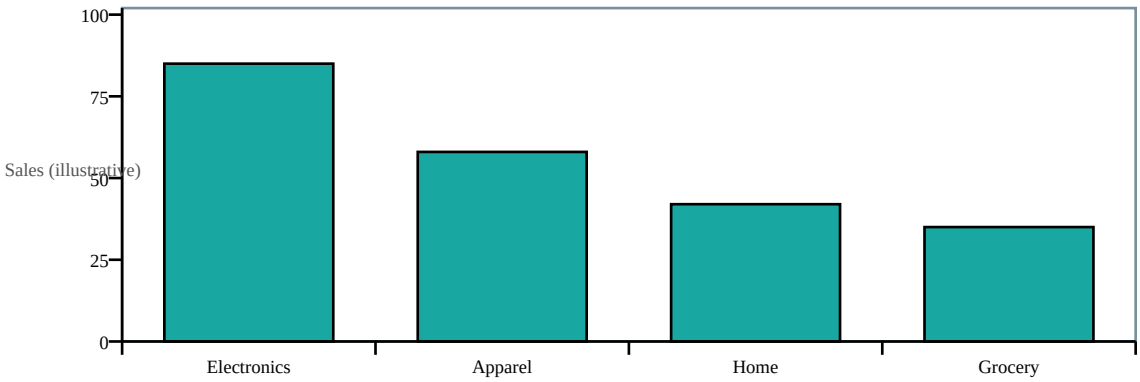
6. Select Home > Publish, choose a workspace, and verify the report in Power BI service.

SAMPLE SOLUTION / EXPECTED OUTPUT

Use these DAX expressions (adjust column names if the workbook differs):

DAX item	Formula
Total Sales	Total Sales = SUM(Sales[Sales Amount])
Total Quantity	Total Quantity = SUM(Sales[Quantity])
Average Order Value	Average Order Value = DIVIDE([Total Sales], DISTINCTCOUNT(Sales[OrderID]))
Calculated column	Sales per Unit = DIVIDE(Sales[Sales Amount], Sales[Quantity])

Illustrative Total Sales by Product Category



EXPECTED OUTCOME

Relationships display 1-to-many markers; measures and calculated column return values; slicers update visuals; the published report opens for users with permission.

VIVA-VOCE QUESTIONS WITH ANSWERS

Q. Why prefer one-to-many relationships from dimension to fact?
Answer: One product or store can relate to many sales rows, supporting consistent filtering.

Q. Measure vs calculated column?
Answer: A measure is evaluated in filter context; a calculated column is computed row by row and stored in the model.

Q. Why use DIVIDE instead of /?
Answer: DIVIDE can handle a zero or blank denominator more safely.

Q. Which visual shows a trend over time?
Answer: A line chart.

Q. What is required for another person to view a published report?
Answer: Publish it to a workspace and grant the appropriate access/licence permissions.

Record note: The output examples are illustrative layouts. Run the practical with the supplied RetailPulse data to obtain your actual values.

Practical 9: Time Intelligence and Advanced DAX for RetailPulse

Related topic: Advanced Data Modeling, Time Intelligence, Complex DAX and Performance Optimization

OBJECTIVE

Create a date table, build time-intelligence and context-modifying measures, and use Performance Analyzer to identify slow visuals.

MATERIALS REQUIRED

Power BI Desktop and the RetailPulse_Analysis.pbix file from Practical 8; at least two calendar years of Sales data for year-on-year measures.

STEP-BY-STEP PROCEDURE

1. Create Date table: Date = CALENDAR(MIN(Sales[Sales Date]), MAX(Sales[Sales Date])). Add Year, Month Number, and Month Name.

2. Mark Date as a date table and relate Date[Date] to Sales[Sales Date].

3. Create Sales YTD and Sales LY measures.

4. Create YoY Growth % with VAR and RETURN.

5. Create Region Share % using CALCULATE and ALL.

6. Build a matrix with Year and Month rows and Total Sales, Sales YTD, Sales LY, and YoY Growth % values.

7. Open View > Performance Analyzer, start recording, refresh visuals, note durations, optimize unused columns/visuals, and record again.

SAMPLE SOLUTION / EXPECTED OUTPUT

Suggested DAX:

Measure	DAX formula
Sales YTD	Sales YTD = TOTALYTD([Total Sales], 'Date'[Date])
Sales LY	Sales LY = CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))
YoY Growth %	YoY Growth % = VAR Curr = [Total Sales] VAR Prev = [Sales LY] RETURN DIVIDE(Curr - Prev, Prev)
Region Share %	Region Share % = DIVIDE([Total Sales], CALCULATE([Total Sales], ALL(Store[Region])))

Year	Month	Total Sales	Sales YTD	Sales LY	YoY Growth %
2024	Jan	…	…	Blank	Blank
2024	Feb	…	…	Blank	Blank
2025	Jan	…	…	2024 Jan value	…

EXPECTED OUTCOME

YTD resets at the beginning of each year; Sales LY reflects the previous year's comparable period; growth is calculated correctly; region shares sum to 100%; Performance Analyzer records visual timings.

VIVA-VOCE QUESTIONS WITH ANSWERS

Q. Why use a continuous date table?
Answer: Time-intelligence functions need a complete, correctly related date sequence.

Q. What does CALCULATE do?
Answer: It evaluates an expression under a modified filter context.

Q. Why use VAR?
Answer: Variables make complex formulas easier to read and can avoid repeating calculations.

Q. What does ALL remove here?
Answer: ALL(Store[Region]) removes the Region filter from the denominator to calculate share of all regions.

Q. Name two ways to improve a slow report.
Answer: Remove unused columns and reduce unnecessary visuals; also optimize DAX and model relationships.

Record note: The output examples are illustrative layouts. Run the practical with the supplied RetailPulse data to obtain your actual values.

Practical 10: Interactive Dashboard, Row-Level Security and Power Platform Integration

Related topic: Interactive Dashboards, RLS, Power Apps and Power Automate

OBJECTIVE

Build interactive report features, restrict regional data using Row-Level Security, and connect Power BI with Power Automate and Power Apps.

MATERIALS REQUIRED

Power BI Desktop, file from Practical 9, a Power BI service workspace with required licence, Power Automate/Power Apps access, and two test accounts. If licensing is unavailable, instructor demonstration may be needed.

STEP-BY-STEP PROCEDURE

1. Add Year and Region slicers; create a Store Detail drill-through page using Store Name; add bookmarks and buttons for chart/table views.

2. Open Modeling > Manage roles and create North Manager and South Manager roles with filters on Store.

3. Use View as to test each role and confirm each sees only the relevant region.

4. Publish the report; in Power BI service, assign test accounts to roles and pin the Total Sales card to a dashboard.

5. Set a data alert threshold and create a Power Automate flow using the data-driven alert trigger to send email or Teams notification.

6. Add a Power Apps visual to Store Detail, pass Store Name and Sales Amount, and create an app to display the selected store's details.

SAMPLE SOLUTION / EXPECTED OUTPUT

Suggested role filters: North Manager: Store[Region] = “North”; South Manager: Store[Region] = “South”. Illustrative interaction map:

Feature	Expected behaviour
Year / Region slicers	Filter report visuals
Drill-through: Store Detail	Opens detail page for selected store
Bookmarks	Switch between chart view and table view
North Manager role	Store[Region] = "North"
South Manager role	Store[Region] = "South"
Power Automate	Alert trigger → email/Teams notification
Power Apps visual	Displays selected Store Name and Sales Amount

EXPECTED OUTCOME

Bookmarks, slicers, and drill-through work; RLS restricts regional users; alert notifications work when the threshold is crossed; the Power Apps visual updates for the selected store.

VIVA-VOCE QUESTIONS WITH ANSWERS

Q. What is Row-Level Security?
Answer: A model security feature that restricts rows visible to users through defined roles and filters.

Q. Why might workspace Admin/Contributor not be restricted by RLS?
Answer: Those workspace roles have elevated permissions and are generally not constrained by RLS in the same way as report consumers.

Q. Drill-through vs drill-down?
Answer: Drill-through navigates to a detail page filtered for a selected item; drill-down explores deeper levels within a visual hierarchy.

Q. What does Power Automate add to a data alert?
Answer: It can trigger an automated action such as sending an email or Teams notification.

Q. Give a business use for a Power Apps visual.
Answer: Let users view or update store/customer details from a report, subject to app and data permissions.

Record note: The output examples are illustrative layouts. Run the practical with the supplied RetailPulse data to obtain your actual values.